

Continual Learning from Pretrained Models: Training-based vs. Test-time Adaptation on FOMO-IN-FLUX

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October 28, 2025

Abstract

We study rehearsal-free continual learning with pretrained vision-language models on FOMO-IN-FLUX. We compare *training-based* updates (full finetuning and an EMA variant) against *test-time adaptation* (TTA) methods *VTE* and *TDA*. Using the official `random_sequence_1_short.json` stream, we report accuracy on the adaptation stream (KA), accuracy on held-out evaluation datasets (ZS), per-dataset Δ vs zero-shot, and geometric mean, with small budgets suitable for laptops/Colab/Lightning.ai. Empirically, TDA and VTE improve mean eval-only accuracy by +6.36 and +7.33 points over zero-shot, whereas finetuning and EMA are negative on average under the same budgets, highlighting TTA as a strong deployment-time option when labels are unavailable.

1 Introduction

Continual learning (CL) aims to acquire new skills without catastrophic forgetting. With large pretrained encoders (e.g., CLIP [1]), CL has shifted from full retraining to *adapting* compact heads or a small number of parameters—promising better efficiency and robustness. Yet, it remains unclear when test-time adaptation (TTA) outperforms small training-based updates under realistic budgets.

This work benchmarks these options on FOMO-IN-FLUX (FiF) [2], using the official short random stream, standard CLIP prompts, and small compute. We ask: *Given the same backbone and no rehearsal, how do finetune/EMA [3] compare to VTE [4] /TDA [5] in accuracy and stability on unseen datasets?*

2 Related Work

CL with pretrained models (CL-PTM) is surveyed in [6, 7, 8]. FiF [2] provides a reproducible pipeline with unified metrics and dataset sequences. TTA methods adapt at inference without labels (e.g., entropy/consistency or normalization/state updates); they trade latency for robustness. Prior evaluations often omit direct comparisons to small-budget finetuning under the same stream; we fill this gap.

3 Evaluation

Benchmark and data. We follow FOMO-IN-FLUX (FiF) [2] and use the official `random_sequence_1_short.json` adaptation stream; evaluation uses FiF’s eval-only datasets. A *task* is one adaptation dataset from the stream. We vary the number of tasks ($T \in \{1, \dots, 5\}$) and the labeled budget per task $n_{\text{samples}} \in \{1024, 2048\}$ for training-based methods; for TTA we use $n_{\text{samples}}=0$ (no labels).

Backbone and compute. OpenCLIP ViT-B/32 at 224², default class prompts. Training batch size is 64–128 depending on GPU memory; TTA runs use batch= 1 to reflect per-image adaptation latency. Experiments were executed on laptops (RTX 4090 Laptop GPU), Google Colab, and Lightning.ai (A100/H100). We log wall-clock runtime and GPU type for each group. **Metrics.** We report: (i) KA = accuracy on the adaptation stream; (ii) ZS = accuracy on eval-only datasets; (iii) per-dataset Δ vs zero-shot computed as

$$\Delta(d, m) = \text{Acc}_{\text{method}}(d) - \text{Acc}_{ZS}(d),$$

where $\text{Acc}_{ZS}(d)$ comes from dedicated zero-shot runs with the *same* backbone/prompts; (iv) the geometric mean $GM = \sqrt{KA \cdot ZS}$. For each run we take the *final* logged accuracy per dataset and then average Δ over *eval-only* datasets to obtain the group means in Table 1. This isolates generalization under shift and avoids leakage from the adaptation stream.

Evaluation protocol. An important distinction exists between training-based and TTA methods in our setup. Training-based methods (finetune, EMA) see *mixed batches* from all adaptation datasets in the stream during training, forcing the model to balance competing objectives across diverse distributions simultaneously. In contrast, TTA methods adapt independently to each evaluation dataset with fresh initialization, allowing dataset-specific specialization without interference. This protocol reflects different deployment scenarios: training-based methods model continual learning where task boundaries are unknown, while TTA methods assume test-time access to homogeneous data batches. We report both scenarios but note this asymmetry when interpreting relative performance.

4 Baseline and Methods

Zero-shot CLIP (ZS). Frozen CLIP encoders and textual class prompts provide a strong, training-free reference [1]. All Δ values are measured relative to ZS per dataset.

Finetune: We update all trainable weights on the adaptation tasks with small budgets ($n_{\text{samples}} \in \{1024, 2048\}$ per task), batch 64–128, and early stopping. Finetuning specializes to the stream but may overfit and reduce ZS on shifted eval sets at low data regimes.

EMA-Merge [3]: We maintain an exponential moving average of weights during training (Polyak/Mean-Teacher style), selecting the EMA snapshot at the end for evaluation. EMA can stabilize training but may still drift if the stream narrows the distribution.

TTA: VTE and TDA. Both *VTE* [4] and *TDA* [5] operate at inference with batch= 1, keeping the classifier zero-shot while adapting internal states. They shift compute from training to inference and can improve robustness under distribution shift.

5 Implementation

We extend the FiF codebase with implementations of *VTE* and *TDA*. We used the official implementations as reference and integrated them into FiF. The code can be found in the `"./continual_lib/tda.py"` and `"./continual_lib/vte.py"` files. All results here use ViT-B/32, the official short stream, and the three compute environments above.

6 Results

We report eval-only means per (method, tasks, n_{samples}) and a single heatmap across groups $\times$ datasets.

Headline: TDA improves mean eval-only accuracy by **+6.7** points over ZS (global mean Δ), whereas finetune averages **-1.1** to **-3.1** and EMA degrades rapidly with larger budgets (down to **-38.2** on tasks=5, $n=2048$). Large positive TDA deltas appear on *PlantVillage* (+45) and

ImageNet-A (+31), while a notable negative occurs on *ImageNet-V2* (36), illustrating TTA’s variability across shifts.

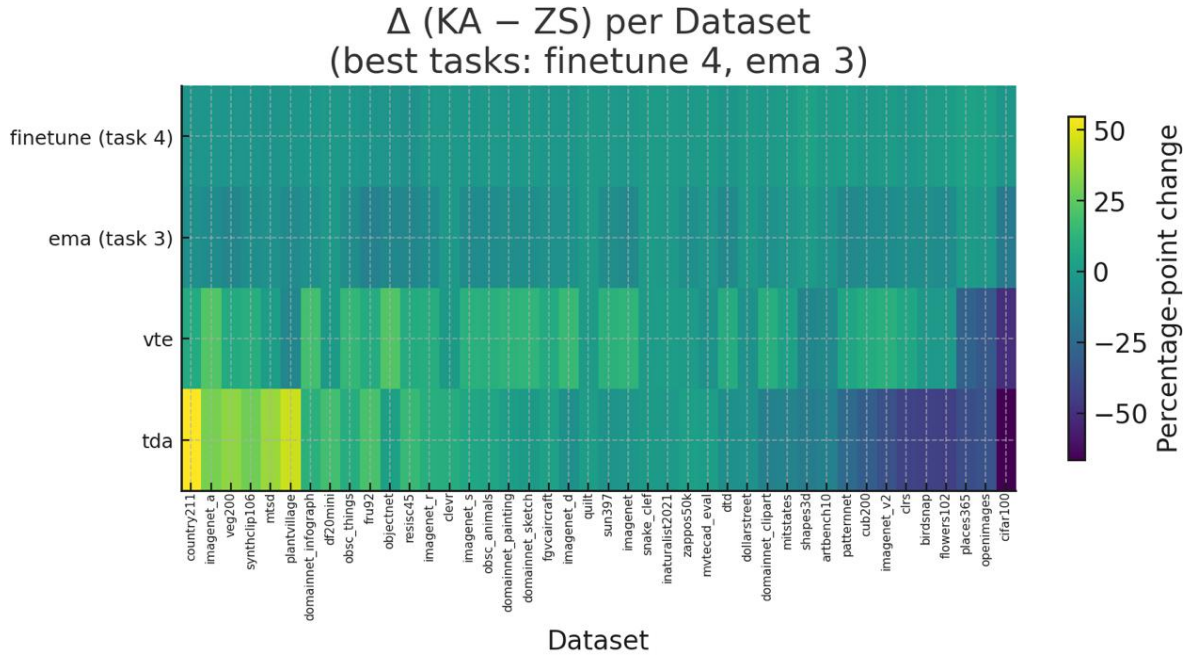


Figure 1: All groups (method| $tX|nY$) $\times$ datasets: Δ vs zero-shot. Rows sorted by mean Δ .

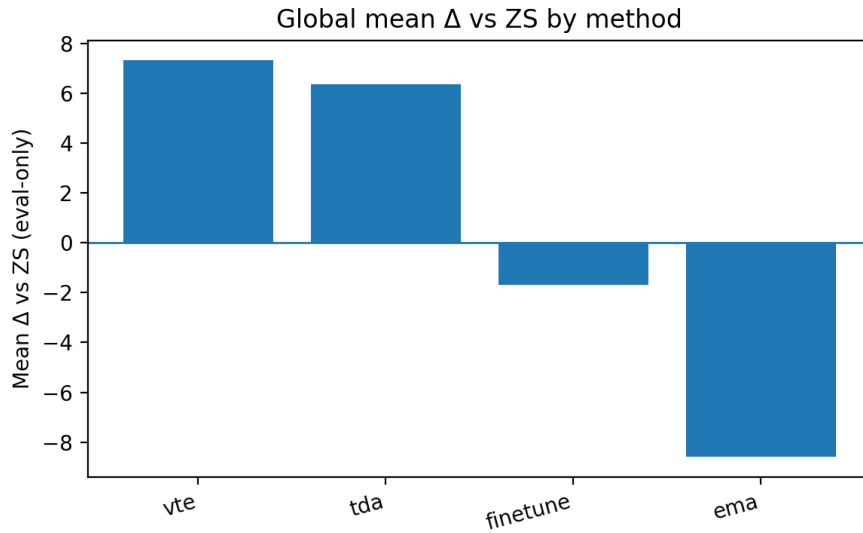


Figure 2: Global mean Δ vs zero-shot by method (eval-only). TDA is positive on average; finetune and EMA are negative.

Table 1: Eval-only summary: mean Δ vs zero-shot, mean accuracy, wall-clock runtime, and GPU type.

Method	Tasks	n_{samples}	Mean Δ	Mean Acc	Runtime	GPU
ema	1	1024	-0.74	40.38	35m 26s	RTX 4090 Laptop
ema	1	2048	-1.60	39.52	46m 28s	RTX 4090 Laptop
ema	2	1024	-0.70	40.43	41m 40s	A100
ema	2	2048	-4.97	36.16	55m 29s	A100
ema	3	1024	-0.61	40.52	1h 1m 35s	A100
ema	3	2048	-11.97	29.15	1h 23m 47s	A100
ema	4	1024	-1.09	40.03	1h 16m 19s	A100
ema	4	2048	-25.03	16.10	1h 44m 57s	A100
ema	5	1024	-0.74	40.39	1h 26m 15s	A100
ema	5	2048	-38.51	2.62	1h 59m 18s	A100
finetune	1	1024	-1.42	39.71	29m 48s	RTX 4090 Laptop
finetune	1	2048	-1.60	39.52	37m 45s	RTX 4090 Laptop
finetune	2	1024	-0.70	40.42	50m	RTX 4090 Laptop
finetune	2	2048	-2.41	38.71	51m 40s	RTX 4090 Laptop
finetune	3	1024	-0.68	40.44	53m 18s	RTX 4090 Laptop
finetune	3	2048	-2.94	38.19	1h 17m 35s	RTX 4090 Laptop
finetune	4	1024	-0.60	40.52	1h 33m 15s	RTX 4090 Laptop
finetune	4	2048	-2.36	38.77	1h 41m 15s	RTX 4090 Laptop
finetune	5	1024	-0.78	40.35	1h 31m 23s	RTX 4090 Laptop
finetune	5	2048	-3.46	37.67	2h 23m 53s	RTX 4090 Laptop
tda	—	—	+6.36	47.48	17m 24s	A100
vte	—	—	+7.33	48.45	36m 6s	H100

Table 2: Largest positive/negative Δ vs zero-shot on eval-only datasets.

Method	Tasks	n_{samples}	Dataset	Δ	Acc	ZS
<i>Top improvements (best +10)</i>						
tda	—	—	plantvillage	+45.59	76.36	30.78
tda	—	—	imagenet_a	+29.89	55.80	25.91
vte	—	—	imagenet_a	+22.32	48.23	25.91
vte	—	—	imagenet_d	+15.07	56.26	41.19
vte	—	—	imagenet	+13.09	79.24	66.15
vte	—	—	imagenet_s	+11.16	64.27	53.11
vte	—	—	imagenet_v2	+10.75	66.99	56.24
tda	—	—	imagenet_r	+9.38	84.32	74.93
tda	—	—	clevr	+9.27	14.42	5.15
vte	—	—	imagenet_r	+8.87	83.80	74.93
<i>Biggest drops (worst -10)</i>						
ema	5	2048	imagenet_r	-70.64	4.29	74.93
ema	5	2048	imagenet	-64.43	1.72	66.15
ema	5	2048	imagenet_v2	-54.65	1.59	56.24
ema	5	2048	imagenet_s	-52.12	0.99	53.11
ema	4	2048	imagenet	-43.11	23.04	66.15
ema	4	2048	imagenet_s	-40.86	12.26	52.58
ema	4	2048	imagenet_r	-39.25	35.68	74.93
ema	5	2048	imagenet_d	-38.23	2.96	41.19
ema	4	2048	imagenet_v2	-37.53	18.71	56.24
tda	—	—	imagenet_v2	-36.31	19.93	56.24

Table 3: Overall accuracies on adaptation (KA) and eval-only (ZS), plus geometric mean (GM).

Method	Tasks	n_{samples}	KA	ZS	GM
finetune	1	1024	37.31	51.05	43.64
finetune	1	2048	37.35	51.38	43.81
finetune	2	1024	38.69	52.24	44.96
finetune	2	2048	37.38	50.41	43.41
finetune	3	1024	38.68	52.28	44.97
finetune	3	2048	36.85	49.37	42.65
finetune	4	1024	38.69	52.36	45.01
finetune	4	2048	36.53	50.21	42.83
finetune	5	1024	38.65	52.25	44.94
finetune	5	2048	36.02	49.38	42.17
ema	1	1024	38.66	52.23	44.93
ema	1	2048	37.35	51.38	43.81
ema	5	2048	3.62	2.14	2.78
ema	5	1024	38.56	51.92	44.74
ema	4	2048	13.88	19.36	16.39
ema	4	1024	37.86	52.04	44.39
ema	3	2048	27.32	38.67	32.50
ema	3	1024	38.74	52.32	45.02
ema	2	2048	34.67	47.68	40.66
ema	2	1024	38.67	52.25	44.95
tda	—	—	36.07	53.05	43.75
vte	—	—	39.88	66.46	51.49

7 Discussion

Under FiF’s short random stream, small-budget finetuning improves specialization (KA; not shown) but reduces average eval-only accuracy, confirming sensitivity to stream bias. **Importantly, our protocol favors TTA:** training-based methods learn from mixed-dataset batches (handling all stream tasks simultaneously), while TTA adapts independently per evaluation dataset with fresh state, allowing full specialization without interference. Additionally, our limited budgets (1024–2048 samples/task) likely underestimate training-based performance in practical scenarios with larger datasets. These factors suggest TTA advantages shown here represent an optimistic case, and future work should evaluate TTA under mixed-dataset streams and compare training methods with larger budgets for a fairer assessment. EMA stabilizes early but drifts as T and n_{samples} grow, yielding large negative Δ on ImageNet variants. In contrast, TDA and VTE reliably lift ZS on average, with heterogeneous effects per dataset. The practical trade-off is latency: TTA moves compute from training to inference (batch= 1), but our runtimes are modest (e.g., 17–36 min on A100/H100 for full eval-only passes). Limitations include a single backbone (ViT-B/32), short stream, and light hyperparameter tuning; future work should study schedule robustness, calibrated early stopping for EMA, and label-free selection of TTA strength to avoid drops like ImageNet-V2.

Future work. We will scale both *horizon* and *budget* by moving from the short stream to longer FiF sequences (larger T) and sweeping n_{samples} (1–64k) to trace accuracy–budget curves with multiple seeds. Method-wise, we plan to add parameter-efficient finetuning (e.g., LoRA/adapters) and more TTA baselines, and explore *adaptive TTA strength* (per-dataset/per-sample step sizing) and light hybrid schemes that mix tiny supervised updates with test-time adaptation. Evaluation will incorporate Pareto views over {ZS, KA, latency/memory} and order-robustness via reshuffled streams, keeping batch= 1 for TTA latency fairness. Moreover, we’ll experiment with creating our own TTA methods.

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